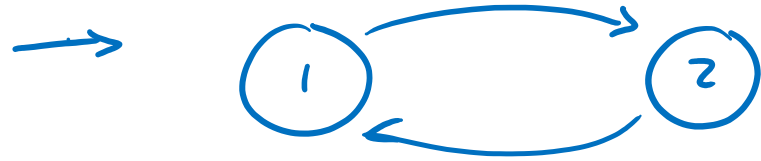


ME 300:
Engineering Thermodynamics

Reversibility



Reversible cycles



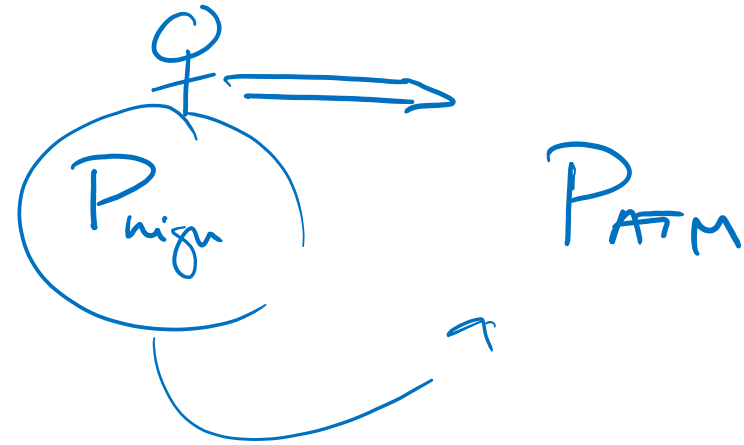
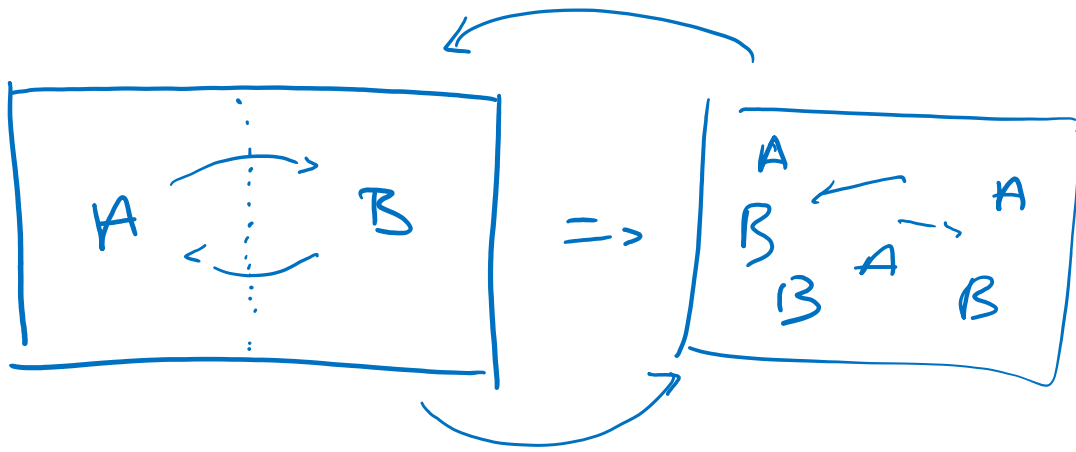
if reversible: sys + surroundings return to original state.

if irreversible: surroundings does not return to original state

=> entropy is generated

→ irreversibilities: friction, mixing,

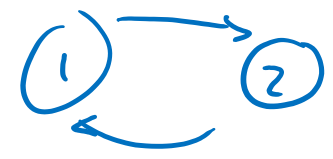
finite expansion => spontaneously



System vs. surroundings

Reversible
cycle

$$\int_1^2 \frac{\delta Q}{T} + \int_2^1 \frac{\delta Q}{T}$$



$$< 0 \Rightarrow \int_1^2 dS' + \int_2^1 dS' = 0$$

1-2 rev
2-1 irrev.

$$\int_1^2 \frac{\delta Q}{T} + \int_2^1 \frac{\delta Q}{T} = \int_1^2 dS_{\text{sys}} + \int_2^1 \frac{\delta Q}{T} < 0$$

= irreversible

$$\Delta S'_{\text{sur}} > 0$$